

House Price Prediction — Project Summary

This project focuses on analyzing housing data and building machine learning models to predict house prices. The dataset was obtained from Kaggle and includes features such as area, number of bedrooms, bathrooms, parking, furnishing status, and other amenities.

In the initial stage, the dataset was explored and cleaned by handling missing values and removing duplicate records. Categorical variables such as furnishing status and yes/no features were converted into numerical format using encoding techniques, making the dataset suitable for machine learning models.

Two models were developed: Linear Regression and Random Forest Regressor. The dataset was split into training and testing sets in an 80:20 ratio. The models were evaluated using performance metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. Among the two, the Random Forest model performed better, indicating its ability to capture complex relationships in the data.

Data visualization techniques were used to better understand the dataset. A histogram showed the distribution of house prices, a correlation heatmap highlighted relationships between variables, and scatter plots demonstrated the relationship between actual and predicted prices.

The analysis revealed that features like area, number of bathrooms, and number of bedrooms have the strongest influence on house prices. One interesting observation was that some features, such as hot water heating, had minimal impact on price.

Based on the findings, it is recommended that real estate businesses focus on properties with larger areas and better amenities, as these significantly contribute to higher property values.

Overall, this project demonstrates how data analysis and machine learning can be effectively used to predict house prices and support decision-making in the real estate sector.